

Technical Report: Bank Data Center Architecture

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Executive Summary

This report presents the design and simulation of a secure, scalable, and highly available bank data center network. The architecture follows a three-tier hierarchical model (Core, Aggregation, Access) and integrates enterprise-grade technologies such as OSPF with MD5 authentication, eBGP for dual ISP connectivity, VLAN segmentation, NAT, QoS, and advanced security systems including firewalls, IDS/IPS, SIEM, and SOC.

The design ensures high availability through redundancy (HSRP and dual-homing), strong traffic control using ACLs and firewalls, and efficient IP utilization via VLSM. The infrastructure simulates a real-world banking environment capable of supporting critical services with high performance and security.

1. Introduction

Banks require highly available, secure, and scalable infrastructure to support financial transactions, internal operations, and online services. Any failure or security breach can lead to severe financial and reputational damage.

This project models a real enterprise bank data center, focusing on reliability, segmentation, and cybersecurity.

2. Global Architecture (3-Tier Design)

2.1 Core Layer

Devices:

- Core-R1, Core-R2 (Routers)
- Core-SW1, Core-SW2 (Layer 3 Switches)

Role:

- Connect to ISPs
- Run eBGP
- High-speed traffic forwarding
- Connect to firewalls

Constraints:

- No direct server connections
 - No heavy filtering (handled by firewalls)
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2.2 Aggregation Layer

Devices:

- Aggr-SW1, Aggr-SW2

Role:

- Inter-VLAN routing (SVIs)
 - OSPF (Area 0 with MD5 authentication)
 - Default gateway using HSRP
 - Connect Access to Core
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2.3 Access Layer

Devices:

- ACC-SW1 to ACC-SW6

Role:

- Connect servers and endpoints
 - VLAN assignment
 - Trunk uplinks to aggregation
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3. Internet and ISP Connectivity

Devices:

- ISP-1 (AS 100)
- ISP-2 (AS 200)

Design:

- Core-R1 ↔ ISP-1
- Core-R2 ↔ ISP-2
- Protocol: eBGP
- Bank AS: 65001

Purpose:

- Redundancy and failover
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4. Firewall Architecture

4.1 FW1 – Perimeter Firewall

Position: Between Core and Aggregation

Functions:

- Block invalid/private IPs
- Allow established traffic
- Protect from external threats

4.2 FW2 – Internal Firewall

Position: Between Aggregation and Server Layer

Functions:

- Control inter-VLAN traffic
 - Example policy: Allow Web → Database (port 3306), deny others
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5. Server Infrastructure

VLAN 20 – Web Servers

- Behind Load Balancer
- HTTP/HTTPS services

VLAN 30 – Database Servers

- Sensitive data storage
- Accessible only from VLAN 20

VLAN 40 – Internal Services

- DNS, DHCP, NTP, VMs

VLAN 50 – SOC

- Security monitoring systems

VLAN 60 – SIEM

- Log collection and analysis

VLAN 70 – Storage

- Backup systems

Security Rack (ACC-SW6)

- SOC Server
 - SIEM Server
 - IDS (SPAN-based)
 - IPS (inline)
 - Load Balancer
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6. Security Systems

IDS

- Passive monitoring via SPAN
- Detects suspicious traffic

IPS

- Inline filtering
- Blocks attacks in real time

SIEM

- Centralized logging
- Detects anomalies (brute force, scans)

SOC

- Human monitoring and response
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7. VLAN Design

VLAN	Purpose
10	Management
20	Web Servers
30	Database
40	Internal
50	SOC
60	SIEM
70	Storage
80	Peering
99	Blackhole

8. IP Addressing (VLSM)

VLAN	Subnet
10	/27
20	/28
30	/29
40	/27
50	/29
60	/29
70	/29

Efficient IP usage reduces waste and improves security.

9. Redundancy Design

- Dual uplinks at every layer
- No single point of failure

Technologies:

- HSRP

Example:

- Virtual IP: 10.0.20.254
 - Aggr-SW1: Active
 - Aggr-SW2: Standby
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10. Routing Architecture

OSPF (Internal)

- Area 0
- MD5 authentication
- Runs on Core and Aggregation

BGP (External)

- Dual ISP peering
 - Failover capability
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11. NAT Configuration

Static NAT

- Public IP → Load Balancer
- Example: 203.0.113.3 → 10.0.20.x

PAT

- Internal users → Internet
 - Example: all → 203.0.113.4
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12. Quality of Service (QoS)

Traffic Type	Priority
Routing	Highest (LLQ)
Banking Apps	40%
Management	20%
Web	25%

13. Access Control Lists (ACLs)

Examples:

- Block private IPs from Internet
 - Allow Web → DB (port 3306)
 - Allow SSH for management
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14. Identified Design Issues and Improvements

- Ensure all devices are dual-homed
 - Correct firewall placement (FW1 between Core and Aggregation)
 - Proper IPS placement (ACC-SW6 inline)
 - Clear VLAN labeling
 - Full HSRP configuration on both aggregation switches
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15. Conclusion

This project demonstrates a complete enterprise-level bank data center architecture. It combines high availability, scalability, and strong security mechanisms to simulate a real-world financial infrastructure. The implementation of redundancy, segmentation, and monitoring ensures resilience against failures and cyber threats.

The final design is suitable for modern banking systems requiring continuous uptime, secure data handling, and efficient network performance.

16. Cisco Configuration Samples

The following samples illustrate key configurations used in the project. IPs and interfaces are representative.

16.1 OSPF with MD5 Authentication (Core / Aggregation)

```
router ospf 1
  router-id 1.1.1.1
  area 0 authentication message-digest
  network 10.0.0.0 0.0.255.255 area 0

interface GigabitEthernet0/0
  ip address 10.0.0.1 255.255.255.252
```

```
ip ospf message-digest-key 1 md5 CISC0123
ip ospf authentication message-digest
```

16.2 eBGP Configuration (Dual ISP)

Core-R1 (to ISP-1 AS 100)

```
router bgp 65001
  bgp log-neighbor-changes
  neighbor 198.51.100.1 remote-as 100
  neighbor 198.51.100.1 description ISP-1
  ! Advertise public prefix
  network 203.0.113.0 mask 255.255.255.0
```

Core-R2 (to ISP-2 AS 200)

```
router bgp 65001
  bgp log-neighbor-changes
  neighbor 192.0.2.1 remote-as 200
  neighbor 192.0.2.1 description ISP-2
  network 203.0.113.0 mask 255.255.255.0
```

16.3 HSRP (Aggregation Layer Default Gateway)

Aggr-SW1 (Active)

```
interface Vlan20
  ip address 10.0.20.1 255.255.255.240
  standby 1 ip 10.0.20.254
  standby 1 priority 110
  standby 1 preempt
```

Aggr-SW2 (Standby)

```
interface Vlan20
  ip address 10.0.20.2 255.255.255.240
  standby 1 ip 10.0.20.254
  standby 1 priority 100
  standby 1 preempt
```


16.4 Access Control Lists (ACLs)

Allow Web to Database (MySQL 3306)

```
ip access-list extended WEB_TO_DB
  permit tcp 10.0.20.0 0.0.0.15 10.0.30.0 0.0.0.7 eq 3306
  deny ip any any
```

Block Private IPs from Internet

```
ip access-list extended BLOCK_PRIVATE
  deny ip 10.0.0.0 0.255.255.255 any
  deny ip 172.16.0.0 0.15.255.255 any
  deny ip 192.168.0.0 0.0.255.255 any
  permit ip any any
```

16.5 NAT Configuration

Static NAT (Public to Load Balancer)

```
ip nat inside source static 10.0.20.10 203.0.113.3

interface GigabitEthernet0/1
  ip nat inside

interface GigabitEthernet0/2
  ip nat outside
```

PAT (Overload for Internal Users)

```
access-list 1 permit 10.0.0.0 0.0.255.255

ip nat inside source list 1 interface GigabitEthernet0/2 overload
```

17. Notes for Implementation

- Ensure all VLAN SVIs participate in HSRP with consistent group numbers
 - Use passive-interface in OSPF for non-routing VLANs
 - Apply ACLs on the correct direction (inbound/outbound) depending on policy
 - Verify BGP failover by shutting one ISP link and testing reachability
 - Monitor logs via SIEM to validate security policies
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18. Server Hardware Recommendations

The following recommendations align with enterprise-grade banking requirements (high availability, performance, and scalability). Equivalent models from other vendors (Dell, Cisco UCS) are acceptable.

18.1 Web Servers (VLAN 20)

Recommended Model: HP ProLiant DL380 Gen10/Gen11

Specs:

- CPU: 2× Intel Xeon Silver/Gold
- RAM: 32–64 GB
- Storage: 2× 480 GB SSD (RAID 1)
- NIC: Dual 10GbE **Notes:** Optimized for load-balanced HTTP/HTTPS services

18.2 Database Servers (VLAN 30)

Recommended Model: HP ProLiant DL380 Gen11

Specs:

- CPU: 2× Intel Xeon Gold
- RAM: 128–256 GB
- Storage: 4–8× SSD (RAID 10)
- NIC: Dual 10–25GbE **Notes:** High I/O performance and redundancy for critical banking data

18.3 Internal Services Server (DNS, DHCP, NTP – VLAN 40)

Recommended Model: HP ProLiant DL360 Gen10

Specs:

- CPU: 1× Intel Xeon Silver
- RAM: 32 GB
- Storage: 2× 480 GB SSD (RAID 1)
- NIC: 1–10GbE **Notes:** Can be virtualized (VMware/Hyper-V)

18.4 SOC Server (VLAN 50)

Recommended Model: Dell PowerEdge R750 / HP DL380

Specs:

- CPU: 2× Xeon Silver/Gold
- RAM: 64 GB
- Storage: 2× 1 TB SSD **Notes:** Supports monitoring dashboards and analyst tools

18.5 SIEM Server (VLAN 60)

Recommended Model: HP ProLiant DL380 Gen11

Specs:

- CPU: 2× Xeon Gold
- RAM: 128 GB
- Storage: 8–12 TB (RAID 5/10) **Notes:** Requires large storage for log retention and analysis

18.6 IDS Server (Detection – SPAN Based)

Recommended Model: HP ProLiant DL360

Specs:

- CPU: Xeon Silver
- RAM: 32–64 GB
- Storage: 1 TB SSD
- NIC: 10GbE (for mirrored traffic) **Notes:** Runs tools like Snort or Suricata

18.7 IPS Server (Prevention – Inline)

Recommended Model: Dedicated appliance or HP DL360

Specs:

- CPU: Xeon Gold
- RAM: 64 GB
- Storage: 1 TB SSD
- NIC: Dual 10GbE (inline) **Notes:** Must handle real-time traffic blocking

18.8 Storage / Backup Server (VLAN 70)

Recommended Model: HP ProLiant DL380 + SAN/NAS (HPE Nimble / Synology Enterprise) **Specs:**

- CPU: Xeon Silver
- RAM: 64 GB
- Storage: 20–50 TB (RAID 6) **Notes:** Used for backups and disaster recovery

18.9 Load Balancer

Recommended Model: F5 BIG-IP VE / NGINX on HP DL360

Specs:

- CPU: 8–16 cores
 - RAM: 16–32 GB
 - Storage: 200 GB SSD **Notes:** Distributes traffic across web servers
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19. Final Recommendation

- Use virtualization (VMware ESXi or Proxmox) to optimize hardware usage
- Implement RAID for all critical servers
- Ensure redundant power supplies for all servers
- Regularly monitor performance and logs via SIEM
- Plan scalability (CPU/RAM upgrades) for future growth